

1.1 Topological Preliminaries

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* A topology T in a set X is a collection of subsets of X (called open sets) satisfying the following axioms:

- (1) the union of any number of open sets is open
- (2) the intersection of any finite number of open sets is open.
- (3) $X, \emptyset$ are open.

*A topological space is a set X with a topology T in X .

$T = \{\emptyset, X\}$ is trivial topology

$T = \mathcal{P}(X)$ is discrete topology

* For $A \subseteq X$, where T is a topology in X . Then T_A is an induced topology in A

$$T_A = \bigcap \{U \setminus A \mid U \in \mathcal{T}\}$$

* In a topological spaces, a closed set is a complement of an open set.

* For $E \subseteq X$, $\overline{E}$ is said to be closure of E , where $\overline{E} = \bigcap V$, S is the set of all closed sets containing E .

* A neighborhood of $x \in X$, is any open set containing x .

* A mapping f from X_1 to X_2 ($X_1 \xrightarrow{f} X_2$) is an assignment of each $x \in X_1$ of an element $f(x) \in X_2$ where X_1, X_2 are topological spaces. (map and mapping are not the same, mapping is a function between two topological spaces).

* The mapping f is continuous if and only if for every open set O_2 in X_2 , the set $f^{-1}(O_2)$ is an open set in X_1 .

* For $x \in X_1$, we say that f is continuous at x , if to every neighborhood U of $f(x)$, there exists a neighborhood

(f is continuous at every point of X_1 .)

We can show the equivalence of both definitions.

Lemma 0.1. Composition of continuous maps are continuous.

Proof. If $f: X \rightarrow Y, g: Y \rightarrow Z$ are continuous.

Goal 1: $(g \circ f)^{-1}(V)$ is open in X where V is any open set in Z .

Goal 2: $(g \circ f)^{-1}(V) = f^{-1} \circ g^{-1}(V)$ $\square$

Example 0.1. (c i) Every mapping is continuous from X to Y if X has discrete topology. (I showed converse is NOT true).

(c ii) If X is a topological space, Y a set with trivial topology, $f: X \rightarrow Y$ is continuous.

* Topological product of X_1, X_2 is $X = X_1 \times X_2$ where topology in X is generated by a basis $\mathcal{B} = \{O_1 \times O_2 \mid O_1 \in \tau_1, O_2 \in \tau_2\}$.

(Defⁿ of basis add to notes, I'm not writing here)

* If $p_1: X_1 \times X_2 \rightarrow X_1$ and $p_2: X_1 \times X_2 \rightarrow X_2$ are such that $p_1(x_1 \times x_2) = x_1$, and $p_2(x_1 \times x_2) = x_2$ resp. Then p_1, p_2 are projections of $X_1 \times X_2$.

Lemma 0.2. Projections are continuous, open mappings.

Proof. It suffice to show for p_1 , and for p_2 is the same argument.

For open map, observe every open set in X can be written as $U \times V$ where U is an open set in X_1 , and V is an open set in X_2 .

Now, $p_1(U \times V) = U$.

For continuity, observe $p_1^{-1}(U) = U \times X_2$ where $U \subset X_1$ is open.

$\square$

Remark 0.1. * A family of subsets $\{V_\alpha\}_{\alpha \in I}$ is a covering if $\bigcup_{\alpha \in I} V_\alpha = X$. If each V_α is an open subset of X , $\{V_\alpha\}_{\alpha \in I}$ is an open covering and if $|I| < +\infty$ then $\{V_\alpha\}_{\alpha \in I}$ is a finite covering.

- A topological space X is **compact**, if every open covering of X contains a finite covering. A subset A is said to be compact, if A is compact in the induced topology.

Remark 0.2. 1. $\mathbb{R}$ is not compact

2. If $A \subseteq \mathbb{R}$, and $|A| < +\infty$, then A is compact.

3. Let $a, b \in \mathbb{R}$, $a \leq b$ then $[a, b]$ is compact in $\mathbb{R}$.

Proof. refer to theorem 27.1 and corollary 27.2 of Munkres.

$\square$

Let X be a simply ordered set having least upper bound property. Then every closed interval in X is compact. (Check out Prahlad sir's notes on General Topology for alternate path)

Given an open covering of $[a, b]$ say, $\mathcal{A}$.

STEP 1: For $x \in [a, b]$ there exists $y \in [a, b]$ and $y > x$ such that $[a, y]$ is covered by at most two elements of $\mathcal{A}$. (Do case where immediate successor exist and immediate successor doesn't exist).

STEP 2: $C = \{y \in [a, b] \mid [a, y] \text{ are covered by finite elements of } \mathcal{A}\}$

Observe $C \neq \emptyset$ by applying STEP 1 for $x = a$ and apply least upper bound property where $c = \sup C$.

STEP 3: Show $c \in C$

STEP 4: $b = c$.

Theorem 0.1. A closed subset of a compact set is compact

Proof. Let X be compact, and Y be a closed subset of X .

For any open covering of Y , say $\mathcal{A}$, we construct $\mathcal{A} \cup \{X \setminus Y\}$ as an open covering of X as $X \setminus Y$ is open. As X is compact, let $\mathcal{A}'$ be a finite subcover of X . Now, the finite subcover of Y we get from $\mathcal{A}'$ will not contain $X \setminus Y$, hence finite subcover of $\mathcal{A}$. $\square$

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Theorem 0.2. A continuous image of a compact set is compact.

Proof. ✓ $\square$

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Theorem 0.3. If X is compact and Y is Hausdorff and $f : X \rightarrow Y$ is continuous, then $f(X)$ is closed in Y .

Proof. ✓ $\square$

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Definition 0.1. A space X is locally compact, if for every $x \in X$, there exists a compact neighborhood of x . *Every compact space is locally compact.*

Proof. ✓ $\square$

• Converse may not be true.

Example 0.2. Example: $\mathbb{R}$

Proof. $\mathbb{R}$ is not compact $\square$

For $x \in \mathbb{R}$, there exists $(x - \varepsilon, x + \varepsilon)$ as a neighborhood of x and $\overline{(x - \varepsilon, x + \varepsilon)} = [x - \varepsilon, x + \varepsilon]$ is compact.

* Homeomorphism is a mapping between two topological spaces X, Y say f such that f is bijective map where f, f^{-1} are continuous mappings.

Example 0.3. Add from C. Chandrashekar.

Note: All topological spaces are T_1 , moving forward unless mentioned otherwise.

* Two sets M_1 and M_2 are separated by open sets if there exist open sets O_1 and O_2 containing M_1 and M_2 such that $O_1 \cap O_2 = \emptyset$.

* Two sets M_1 and M_2 are separated by a real function if there exists a continuous real function $f : X \rightarrow [0, 1]$ such that $f(x) = 0$ for $x \in M_1$ and $f(x) = 1$ for $x \in M_2$.

T_1 : For any disjoint point, there exists an open set containing one and not the other. Equivalently, all singletons are closed.

T_2 (Hausdorff): Two distinct points can be separated by open sets.

T_3 (Regular): For closed set F and $x \notin F$, can be separated by open sets.

T_4 (Normal): Disjoint closed sets can be separated.

- by open sets.

Remark 0.3. $T_4 \Rightarrow T_3 \Rightarrow T_2 \Rightarrow T_1$.

Example 0.4. From the book.

Lemma 0.3. Urysohn's lemma

In a normal space, every two disjoint closed sets can be separated by a real function.

Theorem 0.4. Tychanoff Theorem

Product of compact sets are compact

There are two approaches of proving this theorem

1. Zorn's lemma
2. Transfinite Induction